SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

RSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I A. Tharun Kumar Reddy (192572068) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Tharun.

Assessment Tasks

Scenario-Based Questions

SET 1

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

Scenario - Based Questions (DR):-

Video Streaming Platform Migration:-

- Cloud Architecture & caching:-
  use a Global content Delivery Network (CDN) like AWS cloudfront to cache video chucks close to users, drastically reducing origin load, The core architecture should utilize an object-Storage Service, for hosting video assets.
- Load Balancing & Auto-Scaling:-
  Deploy a layer 7 Application load Balancer (ALB) to route traffic across Mutliple availability zones. Implement target-tracking auto-scaling Policies based on Metric triggers like Cpu utilization or custom Network throughput Metrics, when users traffic spikes additional Compute.
- Cost optimization Strategies:-
  - leverage spot Instances for stateless of transcoding workloads to save up to 90% on-demand Pricing.
  - Configure lifecycle Policies on object storage to Move, older.

(2)

(3,

Fintech Container Deployment Across Multi-cloud:-

- Adopt an Enterprise-Grade kubernetes Management Solution like Red Hat openshift, Google Anthov, (or) Rancher, this Provides a unified control Plane to Manage cluster deployments accross distinct clouds.

Ex:-

AWS and Azure.

- Service Discovery & Scaling:-

Implement a Service Mesh list consul to Manage across-cluster service discovery dynamically. Scaling is handled at two layers, Horizontal Pod Autoscaler (HPA) triggers Pod duplication inside cluster.

- Benefits & Risks:-

Benefits:-

Complete Vendor lock-in Avoidance, optimized Geographic Proximity for users, and High regulatory resilience.

Risks:-

Data Egress fees between cloud environments, Complex Security Configurations, Management, engineering team.

(31)

E-Learning Backup & Disaster Recovery (DR):-

- Cloud-Native DR solution:-

  implement object storage versioning alongside cross-region replication (CRR), For databases, enable automated daily snapshots and continous Point-in-time recovery (PITR). Set up a strick Access-control Policy (IAM).

- Achieving RTO & RPO:-

  RPO:- Recovery Point objective, Met by taking frequent transaction log backups.

  Ex:- Every 5 Minutes.

  RTO:- Recovery Time objectiver, Met by using the-infrasturture-as-code (Iac like Terraform) to Mirror-Environment.

- Business continuity steps:-

1. Detect failure automatically using continuous health probes.
2. Failover traffic Globally at the DNS level-to the disaster recovery region.
3. Promote the replicated read-replica database to Primary Writer.

4. Government Hybrid cloud Model :-

- Secure Hybrid Architecture :-

Establish a dedicated physical Network Connection using a service like AWS Direct connect (or) Azure ExpressRoute combined with an encrypted IPsec, VPN tunnel.

- Identity Management & encryption :-

- Identity :-

Implement federated identity management by connecting the on premise Active Directory to the cloud using SAML 2.0.

- Encryption :-

Enforce Application layer Encryption, utilizing Customer Managed keys (CMKs), inside a dedicated cloud Hardware security Module (HSM).

- Management challenges :-

Maintaining uniform compliance and auditing postures, across differing environments, the handling latent data synchronizations over the physical wired, and Managing split-pane Monitoring interfaces.

(3) High Availabilty Saas provider:-

- Multi-Region strategy:-

Deploy an Active-Active Multi-Region Setup. Global traffic is dynamically routed to the nearest operational region using a Latency-based data-base DNS router.

Exc:-

AWS Route 53 (or) cloudfare Global Traffic-Manager.

Data Replication & failover:-

utilize Global Multi-region databases that support asynchronous cross-region replication with sub-second lag. if a region replication, the DNS-routing health checks instantly isolate the failed region, automatically promoting the secondary region to handle all Gobal.

Monitoring & Alerting:-

Implement synthetic Monitoring agents that a simulate end-user interactions Globally. Set up Predictive anomaliey alerts via Platforms like Datadog (or) AWS Cloudwatch Alarms to flag in Spiking Error rates.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.
Design a solution using cloud-native backup, replication, and versioning.
Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
Some applications must remain on-premises due to compliance requirements.
Design a hybrid architecture that ensures secure communication between environments.
Explain how identity management and encryption should be implemented.
Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
The system must continue operating even if one region goes down.
Propose a multi-region deployment strategy in the cloud.
Explain how data replication and failover mechanisms should be configured.
Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented